
DISTANCE, MOBILITY, AND MODE CHOICE : A SPATIAL ANALYSIS OF SCHOOL TRAVEL BEHAVIOUR IN IRELAND

A PREPRINT

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Abstract

In Ireland, whether a child can walk or cycle to school depends heavily on where they live, something that has been described as a postcode lottery. This study investigates whether the location of schools and how easily they can be reached explains differences in active travel rates across Ireland. We used data from the 2022 Census, the Department of Education, and OpenStreetMap to measure school accessibility for over 16,000 small areas across Ireland. Two methods were used to measure accessibility-kriging, which estimates the density of school places across space, and r5r, which calculates how many school places can be reached within a given walking or cycling time using the real road network. We then used Bayesian Beta regression to model the relationship between accessibility and the proportion of children walking or cycling to school.

Keywords -sss

1 Introduction

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1.1 1.1 Background

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LaTeX command can be used to reference other section. See Section 1.1. However, you can also use **bookdown** extensions mechanism for this.

1.2 1.2 The Postcode Lottery Argument from the article

1.3 1.3 Research Objectives

2 2 Data

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2.2 2.2 Census Data

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3.2 3.2 Travel Time Accessibility:r5r

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4.2 4.2 Walking and Cycling Accessibility: travel time

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4.4 4.4 Model Results — Urban Areas,rural if possible

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